

1. What is Databricks Delta Lake?

Delta Lake is an open-source storage layer developed by [Databricks](#) that sits on top of a data lake and adds reliability features similar to a database.

Features

- **ACID Transactions** (safe concurrent reads/writes)
- **Schema Enforcement** (prevents bad data)
- **Schema Evolution** (allows controlled schema changes)
- **Time Travel** (query older versions of data)
- **Faster Queries** using metadata and optimizations

Example:

```
df.write.format("delta").save("/data/sales")
```

A Delta table contains:

sales/

```
|— part-0001.parquet
|— part-0002.parquet
|— _delta_log/
```

The **_delta_log** folder stores transaction history and versions.

2. Database vs Data Warehouse vs Data Lake vs Delta Lake

Feature	Database	Data Warehouse	Data Lake	Delta Lake
Purpose	Run applications	Business analytics	Store raw data	Reliable data lake analytics
Data Type	Structured	Structured	Any type	Any type
Examples	MySQL, MongoDB	Snowflake	Azure Data Lake Storage	Delta Lake
Updates	Frequent	Rare	File-based	ACID transactions
Analytics	Limited	Excellent	Limited	Excellent
Historical Data	Limited	Yes	Yes	Yes
Versioning	Usually No	Limited	No	Yes
Time Travel	No	No	No	Yes

Database

- Stores current operational data.
- Example: customer orders in an e-commerce app.

Data Warehouse

- Stores cleaned historical data for reporting.
- Example: last 5 years of sales data.

Data Lake

- Stores all raw data (CSV, JSON, images, logs, videos).
- Example: sensor data, application logs, uploaded files.

Delta Lake

- A data lake with database-like reliability.
- Adds transactions, versioning, schema checks, and time travel.